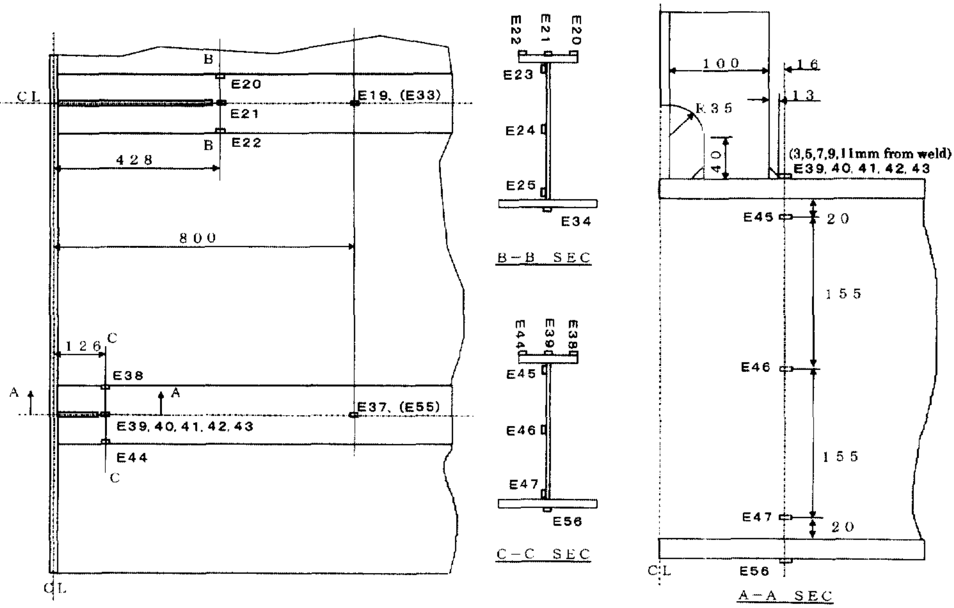

Fig. 2. Locations of strain measurement.

stiffened by flat bars at the side longitudinals. The model is made of high strength hull structural steel of class KA32 having the yield strength higher than 314 MPa (32 kgf/mm^2), and constructed by applying the same welding procedure as used for actual ship structures.

The load is applied in a three-point bending configuration in the horizontal direction, using a hydraulic actuator, through the triangular loading plate of 40 mm thickness attached to the flange plate of the transverse frame. The model was supported at vertical reaction columns through rollers at the ends of longitudinals. While increasing the loads gradually from zero to a maximum of 392 kN (40 tonf), displacements and strains were measured at several reference points and critical locations. Uni-axial strain gauges were used to measure the longitudinal strain components. At the weld joint of the side longitudinal flange plate and the flat bar stiffener, which is expected to be the possible location of fatigue crack initiation, stress concentration gauges were used to evaluate the hot spot stresses. Measuring points, used for the comparison with the analysis results, are shown in Fig. 2.

3. Analysis models

Nine analysis models, shown in Table 1, were collected for this comparative study from seven committee members and one outside collaborator. Mesh subdivisions for some of the models are illustrated in the appendix.